Design of Enterprise IoT

Tutorial 2

Selecting IoT Devices

IoT Use Case: Intelligent Port System for Fraud Detection



Photo credit to Smart Port Solution

Please read the use case and provide a list of essential IoT devices for the proposed solution

Intelligent Port System for Fraud Detection

Port operations play a crucial role in global trade, handling vast amounts of cargo daily. However, fraudulent activities, such as cargo theft, pose significant financial and operational risks. According to China Police Daily, between 2013 and 2017, Chinese Public Security Traffic Management Authorities cracked over 4,000 criminal cases related to port logistics (Song et al., 2020) . Cargo theft is often considered a "silent crime" because losses frequently go unreported and may only be discovered weeks or even months later.

One common cargo theft tactic involves manipulating a truck's empty weight before entering the port. Fraudsters add materials like liquid tanks, cement, or scrap to artificially increase the truck's weight during the initial weigh-in. After getting the empty truck weighed, these materials are secretly removed, creating a weight discrepancy that allows the truck to load excess cargo undetected. Such fraudulent practices lead to financial losses, regulatory violations, and operational inefficiencies.

Traditional fraud detection methods rely heavily on manual inspections, post-event investigations, and rule-based monitoring systems, which struggle to detect anomalies in real time. These limitations highlight the need for an intelligent port system that enhances security, improves efficiency, and minimizes fraudulent activities. IoT technology offers a promising solution by enabling real-time tracking, automated data collection, and intelligent analytics, allowing ports to proactively detect fraud and optimise operations.

Song, R., Cui, W., Vanthienen, J., Huang, L., & Wang, Y. (2022). Business process redesign towards IoT-enabled context-awareness: the case of a Chinese bulk port. *Business Process Management Journal*, *28*(3), 656-683.

Proposed IoT Solution

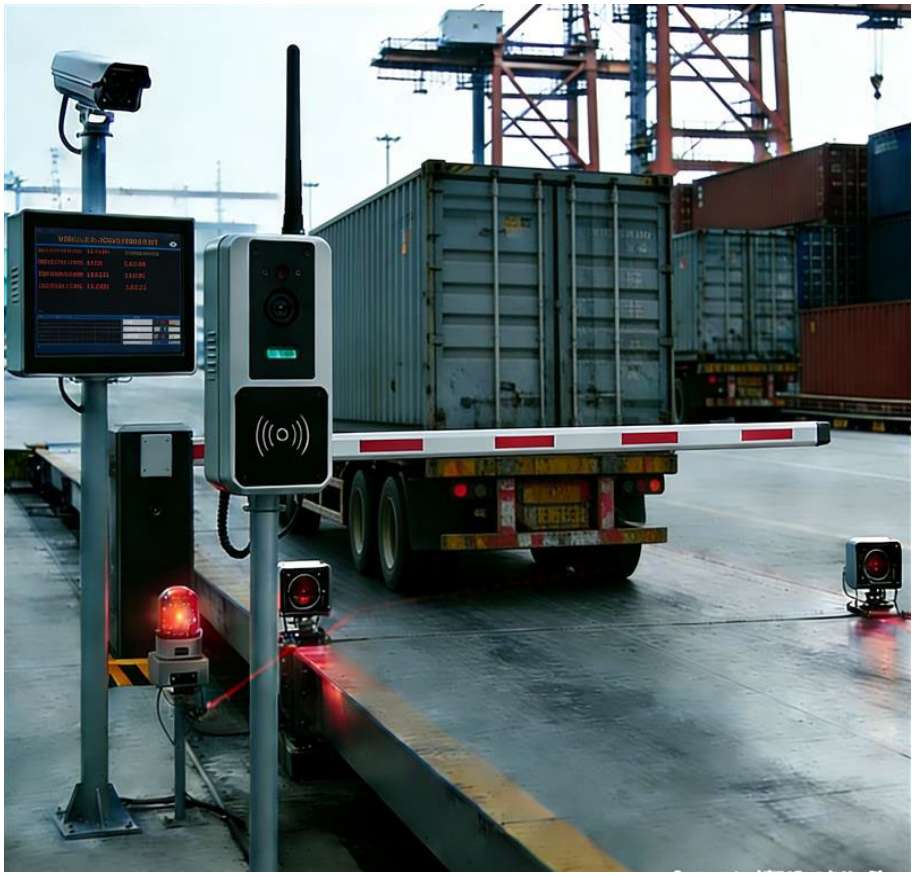
An IoT-based intelligent port system leverages advanced IoT technology and data analytics to enhance security and efficiency. Key components include:

- **Smart Weighbridge Systems:** Infrared sensors and weight sensors detect improper truck positioning and compare real-time weight with historical records, ensuring accuracy in cargo measurement.
- **Environmental Monitoring:** Weather sensors provide real-time rainfall data to detect fraudulent weight manipulation involving hidden water tanks.
- **GPS-Based Route Monitoring:** GPS tracking ensures trucks follow designated routes, detecting unauthorised stops or route deviations that may indicate fraud.
- **IoT-Enabled Cargo Surveillance:** Smart cameras, RFID scanners, and environmental sensors verify cargo handling, preventing unauthorised cargo movement.

Task: Choose and analyse ***off-the-shelf*** devices to be used in the proposed solution (use "***Template-2_IoT-Device-Profile.docx***").

Smart Weighbridge System





Environmental Monitoring



GPS Monitoring



Selecting IoT Devices

What is the device used for? And why this device?

- Feature of Interest
  - What "thing" is being observed and/or acted upon?
    - E.g., vehicle, room, soil, patient, etc.
- Property
  - Which aspect(s) of the "thing" is/are being observed and/or acted upon?
    - E.g., license plate number, temperature, humidity, moisture level, vital sign, etc.
- Characteristics of the device
    - E.g., measurement range, power consumption, environment compatibility, accuracy and precision, response time, data pre-processing capabilities, cost-effectiveness, etc.

Profiling Selected IoT Devices

IoT Device Profile

Name	Type	Feature of Interest	Property	Characteristics	Components	Justification
The name of each IoT device	E.g., sensor, actuator, or a device of both	What "thing" is being observed and/or acted upon?	Which aspect(s) of the "thing" is/are being observed and/or acted upon?	What are the characteristics of the device?	What are the components embedded in the device?	Why is the device selected for the proposed IoT solution?

Source of Information

- Explore **off-the-shelf** IoT devices
  - E.g., Phase IV, AuroRus, Libelium, Garmin sensors, Episensor, InvenSense, etc.
- Check Product/Technical Specification of each device
  - E.g., ZHT-10, HTS-11 Wireless Temp & Humidity Sensor by Episensor https://episensor.com/wp-content/uploads/2022/12/Datasheet-ZHT-1X-EPI-201-00-1.pdf

Extension (NOT covered in this unit):

- *Developer kit and sensors: LoRa rapid development, Intel Galileo (supported sensors), Arduino sensor kit, etc.*